Deep Reinforcement Learning for UAV Placement over Mixed FSO/RF-Based Non-terrestrial Networks

Tinh V. Nguyen*, Hoang D. Le*, Vuong Mai†, Swaminathan R.‡, Anh T. Pham*
* Computer Communications Laboratory, The University of Aizu, Aizuwakamatsu 965-8580, Japan
† Bradford-Renduchintala Centre for Space AI, University of Bradford, BD7 1DP Bradford, U.K.
† Department of Electrical Engineering, Indian Institute of Technology (IIT) Indore

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***Abstract*—Non-terrestrial Network (NTN) architecture, leveraging high-altitude platforms (HAP)-based free-space optical (FSO) backhaul and unmanned aerial vehicles (UAV) for radio frequency (RF) last-mile access, is a promising solution for the future 6G era. Nevertheless, the mobility of the end-users, together with time-varying turbulence/clouds in backhaul links, poses significant challenges in deploying UAVs to maximize the end-to-end network performance. This paper introduces a framework that utilizes deep reinforcement learning (DRL) to optimize UAV placement, considering both dynamic backhaul and access constraints. The results indicate that the trained agent can effectively learn from the environment, confirming its effectiveness in maintaining a relatively high end-to-end throughput performance.**

***Index Terms*—Non-terrestrial networks (NTN), UAV placement, deep reinforcement learning (DRL), Mixed FSO/RF.**

I. Introduction

Recent years have witnessed the growing development of non-terrestrial networks (NTN) in providing the Internet from space [1]. Such networks guarantee cost-effective, seamless, and ubiquitous service availability, especially in remote, hotpot, or emergency areas, where the ground base stations (BSs) are unavailable [2]. The integration of high-altitude platforms (HAP) for data backhaul and unmanned aerial vehicles (UAV) for last-mile access forms the end-to-end air-ground integrated network (AGIN). The AGIN is the promising and pragmatic NTN architecture towards the future 6G era thanks to rapid/flexible deployment and low latency/power consumption compared to satellite systems. To fulfill the ever-increasing demand for high-speed data connections, free-space optical (FSO) technology is a promising candidate for HAP backhaul networks [2]. FSO communication, thanks to its huge unlicensed bandwidth, can provide extremely high link capacity over long distances without depleting radio frequency (RF) resources [3].

The deployment of AGIN, incorporating HAP-based FSO backhaul and UAV-based RF last-mile access, is without challenges. The critical concerns, including pointing misalignment, atmospheric turbulence, and cloud coverage, considerably deteriorate the performance of HAP-based FSO backhaul links. Additionally, the dynamic last-mile access due to the movement of end-users poses various challenges for the deployment of UAV-mounted BS. This, in turn, raises the crucial issue of optimal UAV placement over dynamic conditions from both backhaul (e.g., movement of clouds) and last-mile access (moving end-users) networks. In other words, we need to find the optimal position of UAV-mounted BS to maximize the end-to-end network performance. An efficient algorithm, indeed, is needed to tackle this critical issue. Extensive research endeavors have been steered to discover effective UAV placement algorithms for end-to-end networks comprising FSO backhaul and RF access links. The existing works, however, still leave some open challenges and issues for future research. First, the current studies mainly focus on terrestrial networks and ignore the effects of the FSO channel conditions [4]–[6]. For NTN networks, channel conditions have an essential impact on the backhaul capacity due to cloud coverage and limited power, which, in turn, needs to be considered deeply. As for the second concern, the current works mainly address the UAV placement problem with static networks [4], [5]. This means that when there are any changes in the network environment, the algorithms need to restart from the beginning to find the optimal solution, which might cause significant computation overhead. Therefore, they are not suitable for dynamic environments.

Toward adaptive UAV placement in dynamic end-to-end networks, artificial intelligence/machine learning (AIML) technologies are promising solutions [7]. Reinforcement Learning (RL), a branch of ML, is particularly effective for decision-making problems. Without initial information about the environment (i.e., the network conditions), an RL agent can learn the optimal movement strategy through experience and trial-and-error, aiming to maximize a specified reward function. Deep reinforcement learning (DRL), which leverages deep neural networks (DNN) into RL, can effectively handle large state spaces and high-dimensional input data without requiring significant memory resources compared to conventional RL. Therefore, DRL holds great potential to address the complex environment of the UAV placement problem in dynamic networks.

This paper studies an optimization framework that utilizes DRL for UAV placement, considering both dynamic backhaul and access constraints. The network conditions continuously vary due to the mobility of users and the movement of clouds. We aim to find the optimal placement of the UAV to optimize the end-to-end throughput performance. Extensive simulations confirm that our algorithm converges and the trained agent can move efficiently to attain high throughput performance.

The remainder of the paper is organized as follows. The sys-

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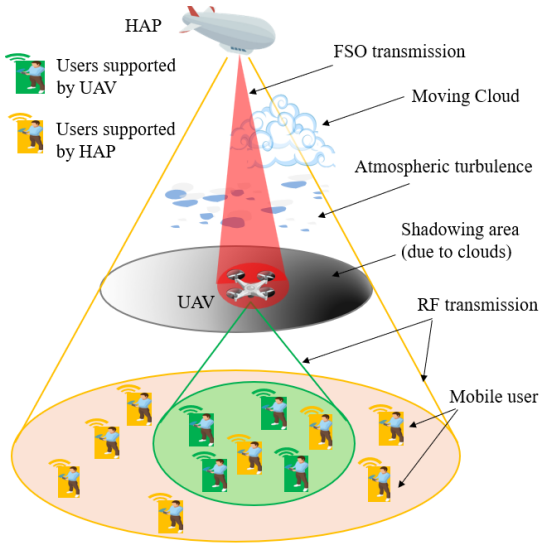


Fig. 1: End-to-end network scenario.

tem and channel models are described in Section II. Section III presents in detail our proposed DRL-based UAV placement algorithm. Numerous numerical results and discussions are given in Section IV. Finally, Section V concludes this paper.

II. System Model

A. System Description

Figure 1 illustrates an end-to-end network scenario comprised of FSO backhaul and RF access links. Specifically, a HAP and a UAV are deployed to provide network services to mobile users (MUs) using RF transmission. At the same time, the HAP provides high-speed backhaul connections to the UAV using FSO transmission. A part of the users are served by the UAV with high speed, while others are served by the HAP with lower speed due to higher loss.

Time is divided into equal time slots. We assume a quasi-static environment, i.e., network conditions stay unchanged during a time slot. As for the FSO backhaul link, we consider a moving cloud with heterogeneous cloud liquid water content (CLWC) values. In addition, the positions of MUs also dynamically change over time. The UAV aims to maximize the total throughput of MUs by finding the optimal position at each time slot, under the constraints of dynamic access networks and limited backhaul capacity.

For convenience, let us denote the index set of the time slots and users as $\mathbb{T} = \{0, 1, ..., T-1\}$ and $\mathbb{U} = \{1, 2, ..., N\}$, where T and N are the total number of time slots and users, respectively. Corresponding, let $t \in \mathbb{T}$ and $i \in \mathbb{U}$ be the index of a particular time slot and user.

B. Data Rate Model of FSO Backhaul Link

1) Channel Model: As for the FSO backhaul link between the HAP and the UAV, we consider three major adverse issues, i.e., cloud attenuation $h_{\rm c}$, atmospheric turbulence $h_{\rm a}$, and pointing misalignment $h_{\rm p}$.

Particularly, the cloud attenuation is widely described by the Beer-Lambert law as $h_{\rm c} = \exp\left(-\sigma_{\rm c} H_{\rm cl} \sec(\xi)\right)$ [8, (11)], where $\sigma_{\rm c}$ is determined by the visibility V which depends on CLWC, as in [8, (12)]. In addition, $H_{\rm cl}$ and ξ are the vertical extent of clouds and zenith angle of the HAP, respectively.

The generalized pointing error model between the HAP and the hovering UAV is well described in [8, Section II-B.3]. The PDF of $h_{\rm p}$ is given as [8, (17)]

$$f_{h_{\rm p}}(h_{\rm p}) = \frac{\varphi_{\rm m}^2}{h_{\rm p}} \left(\frac{h_{\rm p}}{A_{\rm m}}\right)^{\varphi_{\rm m}^2}, 0 \le h_{\rm p} \le A_{\rm m}, \tag{1}$$

where $A_{\rm m}$ and $\varphi_{\rm m}$ can be found in [8, Section II-B.3].

As for the atmospheric turbulence phenomenon, the intensity fluctuations of HAP-to-UAV optical links can be well-described by a Log-normal model, i.e.,

$$f_{h_{\rm a}}(h_{\rm a}) = \frac{1}{h_{\rm a}\sigma_{\rm R}\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{\ln h_{\rm a} + \sigma_{\rm R}^2/2}{\sigma_{\rm R}}\right)^2\right], \tag{2}$$

where $\sigma_{\rm R}^2$ is the Rytov variance determined as in [8, (15)].

Given the composite channel gain $h = h_{\rm a} h_{\rm c} h_{\rm p}$, the instantaneous signal-to-noise ratio (SNR) can be determined as $\gamma = B_0 \overline{\gamma} h^2 = \frac{\left(P_{\rm t}^{\rm FSO} h\right)^2}{\sigma_{\rm n}^2}$, where $\overline{\gamma}$ is the average SNR, $B_0 = 1/\mathbb{E}[h^2]$, $P_{\rm t}^{\rm FSO}$ is transmitted power and $\sigma_{\rm n}^2$ is the receiver noise variance. Utilizing some mathematical operations, the PDF of SNR can be written as [9, (14)]

$$f_\gamma(\gamma) = \frac{\varphi_{\rm m}^2 \gamma^{\varphi_{\rm m}^2/2-1}}{4\left(A_{\rm m} h_{\rm c} \sqrt{B_0 \overline{\gamma}}\right)^{\varphi_{\rm m}^2}} \operatorname{erfc}\left(\frac{\log\left(\frac{1}{A_{\rm m} h_{\rm c}} \sqrt{\frac{\gamma}{B_0 \overline{\gamma}}}\right) + \mu}{\sqrt{2}\sigma_{\rm R}}\right) \times \exp\left(0.5\sigma_{\rm R}^2 \varphi_{\rm m}^2 (1 + \varphi_{\rm m}^2)\right), \tag{3}$$

where $\mu = 0.5\sigma_{\rm R}^2(1 + 2\varphi_{\rm m}^2)$.

2) Ergodic Capacity: The ergodic capacity of the FSO backhaul link can be given as [10]

$$C^{\rm FSO} = \int_0^\infty B^{\rm FSO} \log_2(1+\gamma) f_\gamma(\gamma) {\rm d}\gamma \tag{4}$$

C. Data Rate Model of RF Access Link

The air-to-ground access links are widely modeled as probabilistic Line-of-Sight (LoS) channels. At any given time, a user is only associated with either UAV or HAP. Let us denote the flying platform (FP) associated with user i at time t as $a_i \in \{\text{H, U}\}$, where 'H' and 'U' denote the HAP and UAV, respectively. The path loss models between them for LoS and NLoS links in dB are respectively [5],

$$\eta_i^{\rm LoS}(t) = 20\log\left(\frac{4\pi f_{\rm c} d_i(t)}{c}\right) + \xi^{\rm LoS}, \tag{5}$$

$$\eta_i^{\rm NLoS}(t) = 20\log\left(\frac{4\pi f_{\rm c} d_i(t)}{c}\right) + \xi^{\rm NLoS}, \tag{6}$$

where f_c is the carrier frequency, $\xi_{\rm LoS}$ and $\xi_{\rm NLoS}$ are the average values of excessive path loss in LoS and NLoS, respectively [5]. In addition, $d_i(t)$ is the distance between the FP and user i, given by $d_i(t) = \sqrt{l_i(t)^2 + z_{a_i}(t)^2}$, where

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$l_i(t) = \sqrt{(x_{a_i}(t) - x_i(t))^2 + (y_{a_i}(t) - y_i(t))^2}$ indicates the horizontal distance at time t.

The probability of having LoS connection between user i and its associated FP a_i is given as [5]

$$P_i^{\text{LoS}}(t) = \frac{1}{1 + a\exp\left(-b(\theta_i(t) - a)\right)}, \tag{7}$$

where a and b are environment-related constants, and $\theta_i(t) = \arctan\left(z_{a_i}(t)/l_i(t)\right)$ denotes the elevation angle of user i.

The average path loss between user i and FP a_i is given as

$$\overline{\eta}_i(t) = P_i^{\text{LoS}}(t)\eta^{\text{LoS}}(t) + (1 - P_i^{\text{LoS}}(t))\eta^{\text{NLoS}}(t) \tag{8}$$

The data rate of user i at time t can be approximated using Shannon-Hartley formula as

$$R_i(t) = \frac{B^{\text{RF}}}{N}\log_2\left(1 + \frac{P_{\text{t},a_i}^{\text{RF}} 10^{-\overline{\eta}_i(t)/10}}{N_0}\right), \tag{9}$$

where B^{RF} denotes the total RF allocated bandwidth, $P_{\text{t},a_i}^{\text{RF}}$ indicates the transmitted power of FP a_i, and N_0 is the total noise power of the downlink channel. Note that the channel bandwidth is equally shared among all ground users.

D. User Movement

The user movement can be well-described by the Gauss-Markov mobility model [11]. Specifically, the speed and direction of user i at the t-th time slot depend on those values at the $(t-1)$-th time slot as follows

$$s_i(t) = \alpha s_i(t-1) + (1-\alpha)\overline{s} + \sqrt{1-\alpha^2}s_{n,i}(t-1), \tag{10}$$

$$d_i(t) = \alpha d_i(t-1) + (1-\alpha)\overline{d} + \sqrt{1-\alpha^2}d_{n,i}(t-1), \tag{11}$$

where $s_i(t)$ and $d_i(t)$ are the speed and direction of user i at time step t. Parameters α, $\overline{s}$, $\overline{d}$ indicate the memory level, mean value of speed, and direction, respectively. In addition, $s_{n,i}(t-1)$ and $d_{n,i}(t-1)$ are random variables sampled from Gaussian distributions [11].

The position of user i is updated at each time step as [11]

$$x_i(t) = x_i(t-1) + s_i(t-1)\cos d_i(t-1), \tag{12}$$

$$y_i(t) = y_i(t-1) + s_i(t-1)\sin d_i(t-1), \tag{13}$$

where $x_i(t)$ and $y_i(t)$ are the x and y coordinates of user i at time interval t, respectively.

III. DRL-based UAV Placement

A. Problem Formulation

We aim to maximize the cumulative users' data rate by optimizing the UAV position. Denote $\mathbb{U}_{\text{U}}$ as a set of users served by the UAV, the optimization problem can be formulated as

$$\mathcal{P}: \max_{(x_{\text{U}}(t), y_{\text{U}}(t))} \sum_{t=0}^{t=T-1}\sum_{i=1}^{N} R_i(t), \tag{14a}$$

$$\text{s.t.} \quad (4), (9), \tag{14b}$$

$$\sum_{u\in\mathbb{U}_{\text{U}}} R_u(t) \le C^{\text{FSO}}(t), \forall t \in \mathbb{T}, \tag{14c}$$

$$(x_{\min}, y_{\min}) \le (x_{\text{U}}(t), y_{\text{U}}(t)) \le (x_{\max}, y_{\max}), \tag{14d}$$

where (14c) indicates that the total data rate the UAV can support GUs must not exceed the backhaul capacity, while (14d) illustrates the position constraints of the UAV. The described optimization problem is a non-linear and non-convex problem due to non-linear and non-convex inequality constraints. In addition, network conditions vary over time, which poses difficulty in using classical optimization methods. Therefore, we move to the online DRL-based method, which can learn and handle the real-time and complex environment.

B. User Association - A Greedy Algorithm

In general, when a user is within the coverage of the UAV, it will be served by the UAV with a high-speed connection. However, due to limited backhaul capacity, the UAV can not support all users. Instead, a set of potential users must be chosen to maximize the network performance. To solve this problem, we use the greedy algorithm, i.e., the user with a higher data rate has a higher priority. The algorithm is summarized as follows

- ***Step 1:*** Determine the user set that is within the UAV coverage $\mathbb{U}_{\text{cov}}$, and initialize $\mathbb{U}_{\text{U}} = \{\}$. Construct an array by sorting all users in $\mathbb{U}_{\text{cov}}$ in descending order according to their data rate. Assume that user u' is the first one in the array.
- ***Step 2:*** Associate u' with the UAV, add u' to $\mathbb{U}_{\text{U}}$, and update the available backhaul capacity as $C^{\text{FSO}}(t) = C^{\text{FSO}}(t) - R_{u'}(t)$.
- ***Step 3:*** Select the next user in the array as u' and return to ***Step 2***. This iteration continues until all users in $\mathbb{U}_{\text{cov}}$ are associated with the UAV or the remaining backhaul capacity is not enough for this user.

C. MDP Formulation

To apply the DRL-based algorithm, we need to reformulate $\mathcal{P}$ as a Markov Decision Process (MDP). The state space, action space, and reward function for the UAV placement problem can be defined as follows.

1) State Space: At time slot t, the system state $\boldsymbol{s}_t$, which describes the current network condition, is treated as the input of the neural network to determine the subsequent action. It includes the current location of the UAV, the location of users, and the cloud heatmap. Specifically,

$$\begin{aligned}\boldsymbol{s}_t = [&x_{\text{U}}(t), y_{\text{U}}(t),\\ &x_1(t), y_1(t), x_2(t), y_2(t), ..., x_N(t), y_N(t), \mathbf{v}_{\text{clwc}}],\end{aligned} \tag{15}$$

where $\mathbf{v}_{\text{clwc}}$ is the vector representing the flattened matrix of CLWC grid, which is described in Section IV.

2) Action Space: At each state $\boldsymbol{s}_t$, the UAV selects one action a_t out of five available options from the action space $\mathbb{A} = \{0, 1, 2, 3, 4\}$, Here, actions "0" to "3" represent moving one step East, West, North, and South, while action "4" keeps the UAV position unchanged.

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Algorithm 1 DQN-based UAV Placement

Initialize the replay buffer D with capacity C
Initialize the Q network with random weights $\boldsymbol{\theta}$
Initialize the target $\hat{Q}$ network with weights $\hat{\boldsymbol{\theta}} = \boldsymbol{\theta}$
for each episode **do**
  Initialize the network environment and receive the initial state $\boldsymbol{s}_0$
  for each time slot t **do**
    Obtain action a_t according to the ϵ-greedy policy
    Execute a_t and perform user association algorithm
    Observe reward r_t and next state $\boldsymbol{s}_{t+1}$
    Store transition $(\boldsymbol{s}_t, a_t, r_t, \boldsymbol{s}_{t+1})$ in replay buffer D
    Sample a random minibatch of transitions from D
    Obtain the target Q-value y_i according to (18)
    Perform a gradient descent step on (17) with respect to $\boldsymbol{\theta}$
    Reset $\hat{Q} = Q$ every fixed number of steps
  end for
end for

3) Reward Function: After performing an action in a certain state, the agent will obtain a reward from the environment. This reward function should correlated with the objective function. Let us denote $R_{\mathrm{U}}(t) = \sum_{u \in \mathbb{U}_{\mathrm{U}}} R_u(t)$ as the total users' rate served by UAV and $r_t = \alpha \sqrt{R_{\mathrm{U}}(t) \times C^{\mathrm{FSO}}(t)}$, the immediate reward function is defined as

$$r(\boldsymbol{s}_t, a_t) = \begin{cases} r_t, & R_{\mathrm{U}}(t) \geq R_{\mathrm{thres}}, \\ 1 - r_t, & \text{otherwise}, \end{cases} \tag{16}$$

where α is a scale factor and R_{thres} is the rate threshold.

D. The DRL-based Algorithm

To solve the formulated MDP, we apply the Q-learning algorithm, a model-free, interactive method that uses a Q-function to estimate the expected reward. However, the large state spaces in our problem make it impractical to use a traditional Q-table due to the high memory demands. To overcome this issue, Deep Q-Network (DQN) [12], which leverages a deep neural network (DNN) to represent Q-function, can effectively handle the large state spaces and capture the intricate interactions between actions and states.

In general, the DQN parameters are updated by minimizing the loss function as

$$\mathcal{L}(\boldsymbol{\theta}) = \left(y_t - Q(\boldsymbol{s}_t, a_t; \boldsymbol{\theta})\right)^2, \tag{17}$$

where $\boldsymbol{\theta}$ is the weight matrix of the Q network and $Q(\boldsymbol{s}_t, a_t; \boldsymbol{\theta})$ is the approximated Q-value at timestep t. In addition, y_t is the target Q-value, given by

$$y_t = \begin{cases} r_t, & \text{if episode terminates at step } t+1 \\ r_t + \gamma \max_{a'} \hat{Q}(\boldsymbol{s_t}, a'; \hat{\boldsymbol{\theta}}), & \text{otherwise} \end{cases}, \tag{18}$$

where γ is the discount factor that determines the impact of future rewards and $\hat{\boldsymbol{\theta}}$ is the weight matrix of the target network $\hat{Q}$. The details of the DRL-based UAV placement are summarized in Algorithm 1.

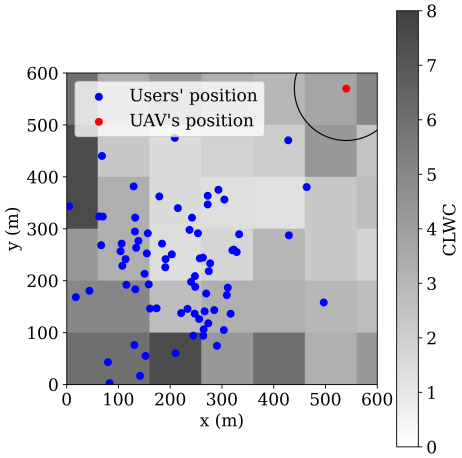


Fig. 2: An example of the environment at time step 10.

IV. Numerical Results

In this section, we conduct extensive simulations to demonstrate the performance of the proposed DRL-based UAV placement algorithm. We consider a 0.6 km $\times$ 0.6 km area with $N = 75$ users normally distributed with a random mean and a standard deviation of 100 m. Twenty percent of users follow the Gauss-Markov mobility model, while the others remain unchanged. The HAP is located at the center of the area, with an altitude of 20 km. The initial position of the UAV is randomly generated during the training process. The UAV has an altitude and a coverage radius of 100 m.

In addition, we consider heterogeneous clouds with different cloud liquid water content (CLWC) values, which are in the range of 0.5 mg to 7.5 mg. In the horizontal plane of the UAV, the projection of the cloud is considered to be a grid, where every 100 m $\times$ 100 m square has a particular CLWC. The cloud moves to the west with a speed of 4 m/s. Note that at a particular time step, we take the average of CLWC values of each 48 m $\times$ 48 m square as the input value of the DQN. Figure 2 illustrates an example of the positions of UAV, ground users, and the CLWC condition at time step 10, where the black circle denotes the coverage of the UAV. Other parameters are as follows.

Backhaul link: transmitted power $P_{\mathrm{t}}^{\mathrm{FSO}} = 6$ dBm, bandwidth $B^{\mathrm{FSO}} = 3$ GHz, HAP's divergence angle $\theta = 1$ mrad, jitter angle $\theta_{\mathrm{jt}} = 50$ μrad, optical wavelength $\lambda = 1.55$ μm, UAV's aperture diameter $D_{\mathrm{r}} = 8$ cm, noise standard deviation $\sigma_{\mathrm{n}} = 10^{-7}$ A/Hz, ground turbulence level $C_{n,i}^2(0) = 10^{-13}$ $\mathrm{m}^{-2/3}$, cloud droplet concentration $N_{\mathrm{c},i} = 250$ cm^{-3}, atmospheric altitude $H_{\mathrm{a}} = 20$ km, Vertical extent of clouds $H_{\mathrm{cl}} = 2$ km, and wind speed $v_{\mathrm{wind}} = 21$ m/s.

Access links: RF transmitted power $P_{\mathrm{t,H}}^{\mathrm{RF}} = 35$ dBm, $P_{\mathrm{t,U}}^{\mathrm{RF}} = 25$ dBm, total bandwidth $B^{\mathrm{RF}} = 300$ MHz, carrier frequency $f_{\mathrm{c}} = 2$ GHz, environment parameters $(a, b) = (9.61, 0.16)$, average excessive pathloss $(\xi_{\mathrm{LoS}}, \xi_{\mathrm{NLoS}}) = (1\text{ dB}, 20\text{ dB})$, and noise power $N_0 = -80.4$ dBm.

DQN model: learning rate $\kappa = 0.001$, discount factor $\gamma =$

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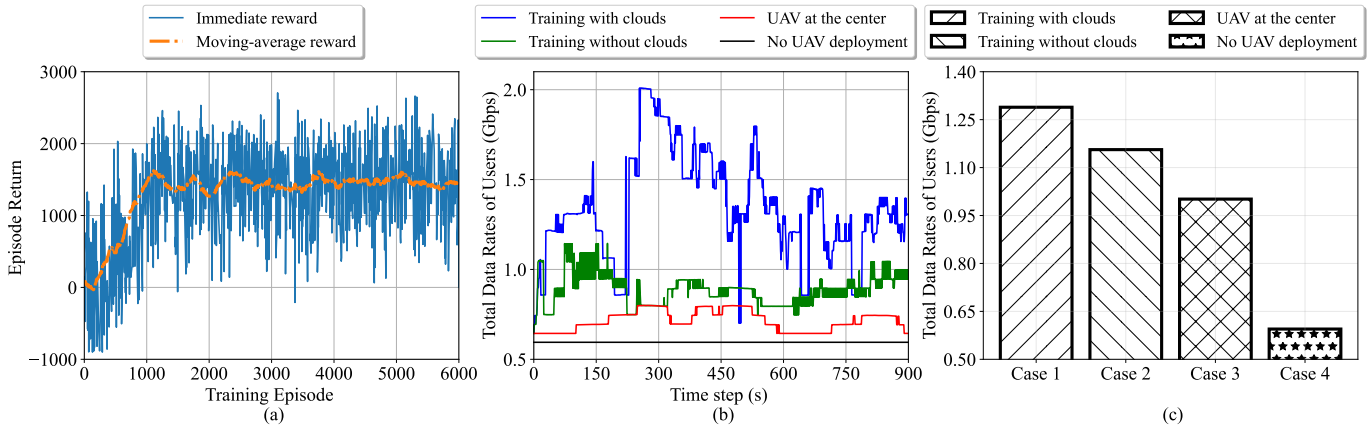


Fig. 3: (a) Episode reward versus training episodes; (b) Total rates of users over time; (c) Average rates of users per time step.

0.99, time slot duration $\Delta t = 1$ sec, total time slot $T = 900$, scale factor $\alpha = 2 \times 10^{-9}$, rate threshold $R_{\text{thres}} = 0.4$ Gbps.

First, we investigate the convergence performance of the algorithm. Figure 3(a) illustrates the immediate and moving-average rewards during the training process. As can be seen, the episode return gradually increases as the number of training episodes progresses and converges after approximately 1200 episodes. This upward trend implies that the agent is learning from the environment and making more efficient decisions to achieve higher returns.

For performance evaluation, we consider other system scenarios, including training the UAV without considering the moving clouds, fixing the UAV at the center of the area, and no UAV deployment. Figure 3(b) demonstrates the total data rates of MUs during 900 seconds (1 test episode). The testing environment is set up randomly, and the UAV has no prior information about this environment. As can be seen, our trained agent, in general, offers the highest performance and can maintain a relatively high data rate over time. This means that the trained agent can move efficiently, leading to significant gaps with other cases. The performance gaps can be clearly seen in Fig. 3(c). In this figure, we illustrate the average data rate of all users at each time step over 100 test episodes. We can see that by considering the moving clouds in the DQN, the throughput of users is increased by approximately 120 Mbps and 300 Mbps, compared to the cases of training without clouds and fixing the UAV at the center of the map, respectively. In addition, by deploying a UAV as a base station, the overall network performance can be considerably improved, which confirms the potential of this architecture for future wireless networks.

V. Conclusion

This paper has presented an optimization framework that utilized DRL for the UAV placement problem. The network conditions continuously vary due to the mobility of users and the movement of clouds. We aimed to find the optimal placement of the UAV to optimize the total throughput of users, under the constraints of both dynamic backhaul and access links. The simulation results confirmed that the UAV can learn well from the dynamic environment and move efficiently to attain high end-to-end throughput performance.

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